The Data Curation Tool within ELIXIR Bio.tools for Health Data Transformation

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The **Data Curation Tool**, developed within the framework of the FAIR4Health project, addresses the challenges posed by data fragmentation and heterogeneity in health research. This **open-source tool**, listed on **Bio.tools**, facilitates the integration of **health data** from various formats and its migration to an **HL7® FHIR® repository**.

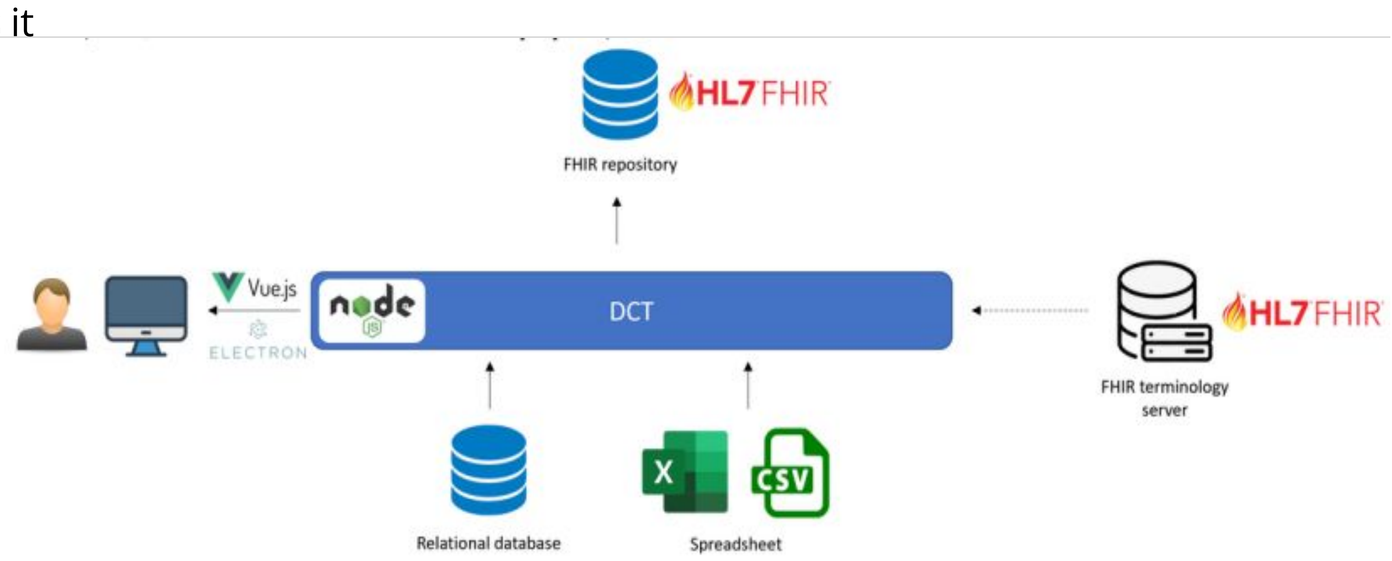
INTRODUCTION

Given the vast amount of **health research data** generated, effective processing is crucial to maximise its usefulness. However, data fragmentation and heterogeneity hinder **efficient analysis and reuse**. To address these challenges, a standalone, desktop application, the **Data Curation Tool (DTC)**, was, developed within the framework of the **FAIR4Health project** (GA/824666) funded by the **European Union's Horizon 2020 programme**. This tool is designed to facilitate the **FAIRification** and **standardization** of **clinical data** in repositories and **health data spaces**, as it ensures that clinical data complies with **FAIR principles** while supporting **federation** and **interoperability** across distributed systems.

It also forms part of a **methodology** to **extract, transform and load (ETL)** existing **health datasets** into **repositories** compliant with **HL7 FHIR**, in alignment with the **FAIR principles**, to standardise and facilitate **health data integration** and **support data treatment**.

SCAN ME!

Methodology



Sinaci AA, Gencturk M, Teoman HA, Laleci Erturkmen GB, Alvarez-Romero C, Martinez-Garcia A, Poblador-Plou B, Carmona-Pírez J, Löbe M, Parra-Calderon CL. A Data Transformation Methodology to Create Findable, Accessible, Interoperable, and Reusable Health Data: Software Design, Development, and Evaluation Study. J Med Internet Res. 2023 Mar 8;25:e42822. doi: 10.2196/42822. PMID: 36884270; PMCID: PMC10034606.

TOOL'S FUNCTIONALITY

The tool, which can be run on **any operating system** (Windows, macOS, Linux), integrates **health data** from **various formats (Excel files, CSV, SQL databases)** and migrates it into an **HL7 FHIR repository**. In addition, it shows users the available **FHIR profiles** to perform the **data mapping** appropriately and can contact a **Terminology Server** to annotate data fields using standard **coding systems** such as **ICD-10** or SNOMED CT

The Data Curation steps include:

- **Verifying** the **FHIR repository** → verify the FHIR repository and connect to a **Terminology Service**
- **Analyzing** the **data source** → browse your file system and **add the sources** as Excel or CSV file
- **Mapping metadata** → map source data with the **FHIR profiles** (fetched from the FHIR repository)
- **Validating** → validate FHIR resources created and **confirm** and **transform** the **data**

ADVANCEMENTS AND FUTURE DIRECTIONS

- ➔ Recent advancements have allowed the tool to **read health data from any server**, extending its functionality beyond the initial **FAIR4Health** repository. With our deep understanding of the tool, we can implement updates to **increase its compatibility with more resources**.
- ➔ This **open-source tool**, included in the web portal developed by **ELIXIR, Bio.tools,** represents a valuable product that will be proposed for inclusion in the **Recommended Interoperability Resources (RIR) FAIR Cookbook,** complying with **ELIXIR Software Management Plan (SMP) requirements.**
- ➔ **The tool's adoption improves** the **integration, standardisation and exchange** of **health data for secondary use**, promoting better collaboration within the scientific community supported by ELIXIR. The tool's performance and usage leads to increased **quality of data**, **efficiency** and **transparency** in research, while helping to **accelerate** the execution of **IA techniques** in the context of health data federations.

Scan me!

Data Curation Tool

Contact & Funding

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